

Principal Component Analysis (PCA)

Geometric Intuition & Core Concepts

Today I started deep diving into **Principal Component Analysis (PCA)** — one of the most important dimensionality reduction techniques in Machine Learning.

This first part focused on understanding the *intuition* before jumping into mathematics.

What is PCA?

PCA is a **dimensionality reduction technique** used to reduce the number of features in a dataset while preserving as much important information as possible.

It transforms original features into a new set of variables called:

Principal Components

These components:

- Are linear combinations of original features
- Are uncorrelated with each other
- Capture maximum variance in descending order

In simple terms, PCA finds new axes that better represent the data.

Why Do We Need PCA?

High-dimensional data often creates problems like:

- Curse of dimensionality
- Increased computational cost
- Overfitting
- Difficulty in visualization

Benefits of PCA:

- Reduces number of features
- Removes multicollinearity
- Speeds up model training
- Helps in data visualization (2D or 3D projection)
- Retains maximum possible information

PCA is especially useful when features are highly correlated.

Geometric Intuition Behind PCA

The most powerful way to understand PCA is geometrically.

Imagine data points scattered in 2D space.

Instead of using the original X and Y axes, PCA:

- Finds a new axis where data spread is maximum
- Rotates the coordinate system
- Projects data onto that new axis

That first new axis is called:

First Principal Component (PC1)

It captures the **maximum variance** in the dataset.

The second axis (PC2):

- Is perpendicular to PC1
- Captures the next highest variance

So PCA is essentially a **rotation of axes** to align with maximum data spread.

What is Variance & Why Is It Important?

Variance measures how spread out the data points are.

In PCA:

- Higher variance means more information
- Lower variance means less contribution

PCA assumes:

The direction with the highest variance carries the most important structure of the data.

So instead of keeping all features, we keep only the components that explain most of the variance.

This is how dimensionality is reduced without losing significant information.